

Name _____ Date _____ Period _____

Graphing $f(x) = |x|$ and Key Attributes

Algebra II | Unit 2 | Day 1 | TEK A2.2.A

Objective: I can graph $f(x) = |x|$ from a table and identify its domain, range, intercepts, symmetry, and minimum value.

Vocabulary

Absolute value — the distance a number is from _____, always _____ (≥ 0).

Vertex — the point where the two straight-line pieces of $f(x) = |x|$ meet: _____.

Key attributes — domain, range, _____, symmetry, and minimum/maximum.

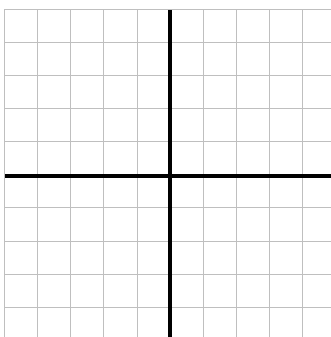
Symmetric about the y-axis — the left and right halves of the graph are _____.

Part 1 — Build the Table, Then Graph

Fill in $f(x) = |x|$ for each x -value.

x	-3	-2	-1	0	1	2	3
$f(x)$							

Plot the seven points and connect them. Each square is 1 unit; both axes run -5 to 5.



Part 2 — Key Attributes of $f(x) = |x|$

Domain: _____

Range: _____

x-intercept: _____ y-intercept: _____

Vertex: _____ Symmetric about: _____

Minimum value: _____ Maximum value: _____

Part 3 — What Does $|x|$ Mean in Real Life? Distance

A hiking trail runs both directions from a trailhead at position 0. A hiker's position is x miles (negative = one direction, positive = the other). The distance from the trailhead is $d(x) = |x|$.

Hiker at $x = -3.5$: distance = _____

Hiker at $x = 6$: distance = _____

A thermometer reads t degrees relative to 0°C (freezing). $|t|$ tells you how many degrees you are from freezing, above or below.

$t = -15$: _____ degrees from freezing $t = 8$: _____ degrees from freezing

In your own words, explain why distance can never come out negative.

Sample: _____

Part 4 — Compare to a Line

Fill in both columns to see what changes and what stays the same.

	$y = x$ (a line)	$f(x) = x $
Domain	_____	_____
Range	_____	_____
x- and y-intercept	_____	_____
Minimum?	_____	_____
Symmetric?	_____	_____

Practice

- $|-8| =$ _____
- $|6| =$ _____
- $|0| =$ _____
- $|-2.5| =$ _____
- If $f(x) = |x|$ and $x = -10$, $f(x) =$ _____
- $f(x) = |x|$ has a vertex at $(0, 0)$. Give one more point on each side of it. Left: _____ Right: _____
- True or False: the graph of $f(x) = |x|$ is symmetric about the x-axis. Explain.
False _____
- The points $(-4, 4)$ and $(4, 4)$ are both on the graph of $f(x) = |x|$. What key attribute does this illustrate?
Symmetry about the y-axis _____
- One point is at $x = -6$, a certain distance from zero. What x-value on the OTHER side of zero is the same distance away? _____
- Which is larger, $|-15|$ or $|9|$? _____
- Put these in order from least to greatest: $|-4|$, $|1|$, $|-9|$, $|6|$

Exit Ticket

For $f(x) = |x|$, give the domain, range, vertex, and axis of symmetry.

Domain _____ Range _____

Vertex _____ Axis of symmetry _____